

K- Theory

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We describe an alternating description of $K(X, A)$ which has particular relevance for products.

If $n \geq 1$ we define $C_n(X, A)$ to be a category as follows: An object of $C_n(X, A)$ is a collection of $E_n, E_{n-1}, \dots, E_0$ of bundles over X together with maps $\alpha_i : E_i|_A \rightarrow E_{i-1}|_A$ such that

$$0 \rightarrow E_n|_A \xrightarrow{\alpha_n} E_{n-1}|_A \cdots \xrightarrow{\alpha_1} E_0|_A \rightarrow 0$$

is exact. The morphism $\phi : E \rightarrow F$ where $E = (E_i, \alpha_i), F = (F_i, \beta_i)$ are collection of maps ϕ_i such that $\beta_i \phi_i = \phi_{i-1} \alpha_i$.

An elementary sequence in $C_n(X, A)$ is a sequence of the form $0, 0, \dots, E_p, E_{p-1}, 0, \dots, 0$ where $E_p = E_{p-1}$.

We define $E \simeq F$ if for some set of *elementary objects* $Q_1, \dots, Q_k, P_1, \dots, P_l$ we have

$$E \oplus Q_1 \oplus \cdots \oplus Q_k \simeq F \oplus P_1 \oplus \cdots \oplus P_l$$

The set of such equivalence classes is denoted by $\mathcal{L}_n(X, A)$. This is a semigroup for each n . There is a natural inclusion $C_n(X, A) \subset C_{n+1}(X, A)$ which induces homomorphism $\mathcal{L}_n(X, A) \rightarrow \mathcal{L}_{n+1}(X, A)$. We denote by $C_\infty(X, A)$ the union of all C_n and by $\mathcal{L}_\infty(X, A)$ the direct limit $\mathcal{L}_n$.

Theorem 1.1. *For all $n \geq 1$, the maps*

$$\mathcal{L}_n(X, A) \rightarrow \mathcal{L}_{n+1}(X, A)$$

are isomorphisms and

$$\mathcal{L}_n(X, A) \simeq K(X, A)$$

Definition 1.2. *An Euler characteristic χ_n for $\mathcal{L}_n$ is a transformation of functors*

$$\chi_n : \mathcal{L}_n(X, A) \rightarrow K(X, A)$$

such that whenever $A = \emptyset, \chi(E_n, E_{n-1}, \dots, E_0) = \sum (-1)^i [E_i]$.

Lemma 1.3. *Let $A \subseteq X$ and let E, F be bundles over X . Let $\phi : E|_A \rightarrow F|_A, \psi : E \rightarrow F$ be monomorphisms (isomorphisms) and assume $\psi|_A \simeq \phi$. Then ϕ extends to X as a monomorphism (isomorphism).*

Lemma 1.4. *If $A = \text{pt}$, then*

$$0 \rightarrow \mathcal{L}_1(X, \text{pt}) \rightarrow \mathcal{L}_1(X) \rightarrow \mathcal{L}_1(\text{pt})$$

is exact. Thus if χ_1 is an Euler characteristic for $\mathcal{L}_1$,

$$\chi_1 : \mathcal{L}_1(X, \text{pt}) \xrightarrow{\sim} K(X, \text{pt})$$

is an isomorphism.

Proof. If (E_1, E_0) represents an element of $\mathcal{L}_1(X)$ whose image in $\mathcal{L}_1(\text{pt})$ is 0, E_1 and E_0 have the same dimension over A . Thus there is an isomorphism $\phi : E_1|_A \rightarrow E_0|_A$.

If (E_1, E_0, ϕ) has image zero in $\mathcal{L}_1(X)$, there is a trivial P and an isomorphism $\psi : E_1 \oplus P \cong E_0 \oplus P$. $\psi(\phi \oplus 1)^{-1}$ is an automorphism of $E_0 \oplus P|_A$. Since A is a point any such automorphism must be homotopic to the identity and hence it extends to $\alpha : E_0 \oplus P \cong E_0 \oplus P$. Thus we have a commuting diagram:

$$\begin{array}{ccc} E_1 \oplus P|_A & \xrightarrow{\phi \oplus 1} & E_0 \oplus P|_A \\ \downarrow \psi|_A & & \downarrow \alpha|_A \\ E_0 \oplus P|_A & \xrightarrow{1} & E_0 \oplus P|_A \end{array}$$

Thus (E_1, E_0, ϕ) represents 0 in $\mathcal{L}_1(X, A)$. □

Lemma 1.5.

$$\mathcal{L}_1(X/A, A/A) \xrightarrow{\sim} \mathcal{L}_1(X, A)$$

is an isomorphism for all (X, A) . Thus if χ_1 is an Euler characteristic,

$$\chi_1 : \mathcal{L}_1(X, A) \cong K(X, A)$$

is an isomorphism for all (X, A)

Lemma 1.6. If χ_1, χ'_1 are two Euler characteristics for $\mathcal{L}_1$, then $\chi_1 = \chi'_1$.

Proof. $\chi'_1 \chi_1^{-1}$ is a transformation of functors from K to itself which is the identity on each $K(X)$. Since $K(X, A) = \tilde{K}(X/A)$ is injected into $K(X/A)$, it is the identity on all $K(X, A)$. □

Theorem 1.7. There exists an Euler characteristic χ_1 for $\mathcal{L}_1$

Proof. Suppose $[E_1, E_0, \alpha] \in \mathcal{L}_1(X, A)$. Let X_0, X_1 be two copies of X and let $Y = X_0 \cup_A X_1$ be the space which results from identifying corresponding points of A . Then $[E_1, \alpha, E_0] \in K(Y)$. Let $r_i : Y \rightarrow X_i$ denote the retractions for $i = 1, 2$. Then

$$K(Y) = K(Y, X_i) \oplus K(X_i)$$

The map $(X_0, A) \rightarrow (Y, X_1)$ induces an isomorphism

$$K(Y, X_1) \rightarrow K(X_0, A)$$

Let $\chi_1(E_1, E_0, \alpha)$ be the image of the component of $[E_1, \alpha, E_0]$ which lies in $K(Y, X_1)$.

If $A = \emptyset$, then

$$\chi(E_1, E_0, \alpha) = [E_0] - [E_1]$$

This definition is independent of the choices made.

Two bundles
each over two
copies of X

$$\begin{array}{ccc} \mathcal{L}_1(X_0, A) & \xrightarrow{\sim} & \mathcal{L}_1(Y, X_1) \\ \downarrow & & \downarrow \\ K(X_0, A) & \xrightarrow{\sim} & K(Y, X_1) \end{array} \quad \begin{array}{c} [E_1, \alpha, E_0] \\ \downarrow \\ \chi_1([E_1, \alpha, E_0]) = \\ [E_1, \alpha, E_0] |_{K(Y, X_1)} \\ \stackrel{A=\emptyset}{=} [E_0] - [E_1] \end{array}$$

□

Corollary 1.8. *The class of (E_1, E_0, α) in $\mathcal{L}_1(X, A)$ only depends on the homotopy class of α .*

Proof. Let $Y = X \times [0, 1]$ and $B = A \times [0, 1]$. Then if α_t is a homotopy with $\alpha_0 = \alpha$, α_t defines $\beta : \pi^*(E_1)|_B \cong \pi^*(E_0)|_B$. Let $i_j : (X, A) \rightarrow (X \times [j], A \times [j])$. From the commuting diagram, every vertical map is an isomorphism and $i_0^*(i_1^*)^{-1}$ is the identity, we have $(E_1, E_0, \alpha_0) = (E_1, E_0, \alpha_1)$.

$$\begin{array}{ccccc}
 (E_1, E_0, \alpha_0) & & & & (E_1, E_0, \alpha_1) \\
 \\
 \mathcal{L}_1(X, A) & \xleftarrow{i_0^*} & \mathcal{L}_1(Y, B) & \xrightarrow{i_1^*} & \mathcal{L}_1(X, A) & (E_1, E_0, \alpha_t) \approx i_t^*(\pi^*(E_1), \pi^*(E_0), \beta) \\
 \downarrow \approx \chi_1 & & \downarrow \approx \chi_1 & & \downarrow \approx \chi_1 \\
 K(X, A) & \xleftarrow{i_0^*} & K(Y, B) & \xrightarrow{i_1^*} & K(X, A) \\
 & \nwarrow & & \nearrow & \\
 [E_1, \alpha_0, E_0] & & i_0^*(i_1^*)^{-1} = \text{id} & & [E_1, \alpha_1, E_0]
 \end{array}$$

□

Lemma 1.9. *The map $\mathcal{L}_n(X, A) \rightarrow \mathcal{L}_{n+1}(X, A)$ is onto for $n \geq 1$*

Proof. If $(E_{n+1}, \dots, E_0; \alpha_{n+1}, \dots, \alpha_1)$ represents an element of $\mathcal{L}_{n+1}(X, A)$, so does

$$(E_{n+1}, E_n \oplus E_{n+1}, E_{n-1} \oplus E_{n+1}, E_{n-2}, \dots, E_0; \alpha_{n+1}, \alpha_n \oplus 1, \dots, \alpha_1).$$

The two maps

$$\alpha_{n+1} \oplus 0 : E_{n+1} \rightarrow E_n \oplus E_{n+1}$$

and

$$0 \oplus 1 : E_{n+1} \rightarrow E_n \oplus E_{n+1}$$

are (linearly) homotopic as monomorphisms. $0 \oplus 1$ extends to X , and thus $\alpha_{n+1} \oplus 0$ extends to a monomorphism $\beta : E_{n+1} \rightarrow E_n \oplus E_{n+1}$ on all of X . Thus we can write $E_n \oplus E_{n+1}$ as $\beta(E_{n+1}) \oplus Q$. Then we see that, if $\gamma : Q \rightarrow E_{n-1} \oplus E_{n+1}$ is the resulting map,

$$(E_{n+1}, \dots, E_0; \alpha_{n+1}, \dots, \alpha_1) \sim (0, 0, E_{n-1} \oplus E_{n+1}, \dots, E_0; 0, \gamma, \dots, \alpha_1)$$

. Thus $\mathcal{L}_n(X, A) \rightarrow \mathcal{L}_{n+1}(X, A)$ is onto.

□

Lemma 1.10. *The map $\mathcal{L}_n(X, A) \rightarrow \mathcal{L}_{n+1}(X, A)$ is an isomorphism for all $n \geq 1$.*

Proof. It suffices to produce a map $\mathcal{L}_{n+1} \rightarrow \mathcal{L}_1$ which is a left inverse of the map $\mathcal{L}_1 \rightarrow \mathcal{L}_{n+1}$.

$$\begin{array}{ccccc} \mathcal{L}_1 & \hookrightarrow & \mathcal{L}_{n-1} & \hookrightarrow & \mathcal{L}_n \\ & & & & \\ & \swarrow & & \searrow & \\ & F & & (0, E) & \\ & \swarrow & \xrightarrow{E} & \searrow & \\ & (0, F) & & & \end{array}$$

Here

$$E = (E_n \dots, E_0; \alpha_n, \dots, \alpha_1) \in \mathcal{L}_n \text{ and } F = (\sum E_{2i+1}, \sum E_{2i}, \sum \alpha_{2i+1} + \sum \alpha'_{2i}) \in \mathcal{L}_1$$

where α' is the Hermitian adjoint of α with respect to a choice of Hermitian metric on E_i . $\square$

Corollary 1.11. *For each n there exists exactly one Euler characteristic $\chi_n : \mathcal{L}_n(X, A) \rightarrow K(X, A)$ and it is always an isomorphism. Thus there exists*

$$\chi : \mathcal{L}_\infty(X, A) \rightarrow K(X, A)$$

isomorphically.